Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/14/2007
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Measurement of Blood and Tissue Chemistry		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01581	**Congressional District:**	3
Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	10/01/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed end date to accommodate FY08 NSBRI submission for final report (jp 5/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heard, Stephen (University of Massachusetts Medical School)		
Grant/Contract No.:	NCC 9-58-SMS00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Trauma and acute medical problems along with loss of skeletal muscle mass, strength and endurance, are some of the most serious risks facing astronauts during long duration spaceflight. The measurement of two metabolic parameters, muscle pH and oxygen can be applied in both areas. This project examines the hypothesis that near infrared spectroscopy (NIRS) is a platform technology that can provide noninvasive physiologic monitoring in support of multiple NASA needs. The immediate goal is to produce, validate and deliver to NASA-JSC a small, lightweight medical monitor which utilizes NIRS to measure important metabolic parameters. Requisite precision and accuracy will be demonstrated for both male and female subjects of any ethnic origin. The device will perform equally well on the forearm and the thigh, despite gender and weight related differences in fat thickness. The specific aims of this project are: (1) Develop calibration procedures for modified NIRS sensor to measure deep muscle metabolic parameters (tissue pH and oxygen); (2) Validate sensor in exercise protocol and deliver systems to NASA-JSC; (3) Determine values for tissue pH and oxygen that can be used by a smart medical system to indicate shock and assist in guiding treatment; and (4) Evaluate hardware for flight requirements and develop plan to produce flight-ready instrumentation. During year 3 of this project we continued to validate and improve the robustness of our calibration equations for noninvasively measuring muscle pH and PO2. We also developed a new methodology for determining muscle oxygen saturation. We completed the development of a system for measuring through thicker fat layers on the thigh and delivered the system to NASA-JSC. As part of this system we developed a mechanical fixture for stabilizing the sensor on the leg and showed that measurements were immune to motion artifacts. A pilot study of 10 subjects was completed at JSC demonstrating performance of our system during a VO2max cycle ergometry test. Using data from our system we demonstrated the capability to accurately determine anaerobic threshold noninvasively from our pH measurements and demonstrated the feasibility of noninvasively determining oxygen uptake using only NIRS-determined parameters. The other intended application of the system will be in a Smart Medical System for the assessment and treatment of critically ill and injured crew. We have begun a study in the UMass Medical School Emergency Department of patients with severe sepsis. We demonstrated that our muscle oxygen saturation measurement was highly correlated with blood lactate, indicating the severity of microvascular impairment and the response of the patient to therapy. We also demonstrated good agreement of our noninvasive pH measurement with pH values determined from venous blood. We have initiated discussions with 3 potential commercial partners who have expressed an interest in working with us to develop miniature hardware which will be more suitable for spaceflight. We have also begun discussions with personnel at NASA Glenn Research Center about assistance in developing hardware which is designed to be flight certified. The NIRS noninvasive metabolic monitor is expected to have many applications for NASA. The system will have additional use on earth for military and civilian personnel treating critically ill and injured patients. It can also be used in the hospital, ambulances and helicopters. As part of a Smart Medical System, advanced medical assessment and monitoring may become available to physicians in remote and rural areas, who may not have access to specialist expertise.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research focuses on developing general techniques that will extend the use of near infrared spectroscopy to multiple analytes and across all human subjects irrespective of their gender or ethnic origin. When applied by us and other researchers we can create new medical diagnostic tests and monitors which can perform accurately without having to draw a blood sample or take a biopsy specimen. Noninvasive measurement on small portable monitors allows more sophisticated medical assessment at the patient bedside, or in an ambulance or emergency helicopter. The sophisticated measurements we propose, muscle pH, PO2 and hematocrit, taken early in shock should initiate more rapid treatment and prevent serious complications like sepsis and multiple organ failure. Noninvasive monitoring allows multiple assessments to be recorded over time without blood loss. In the hospital or ICU, in combination with smart computer intelligence, might allow much earlier indication of serious conditions which must be treated. We have recently shown that the measurement of muscle pH and PO2 can be made on exercising muscle and is indicative of work intensity. We expect that the continuous noninvasive measurement of these parameters might be used to help optimize the rehabilitation of patients with muscle injury or weakness. The technology is also capable of assessing regional perfusion abnormalities. We are currently studying its application as an early indicator of serious risk for the formation of diabetic foot ulcer. If these noninvasive measurements are done as part of routine screening of diabetic patients it might be possible to start preventive treatment in time to prevent ulcers and their complications like amputation.
Task Progress:	There are 3 separate aspects of this program that are being pursued in parallel: 1) enhancement of calibration equation accuracy, 2) development of hardware for leg measurement and 3) application of the system to assess trauma patients and guide treatment. 1) Calibration Equations. We developed a novel method to correct transdermal spectra for interference from overlying layers of skin and fat. This method employs a specially designed sensor and novel mathematics. This methodology was published and is patent pending. We then found that we also needed to correct for variation in muscle optical properties between subjects. We developed a new mathematical method for this and demonstrated it in a phantom study, and on pH measurements for exercising subjects. A paper has on this technique has recently been published in the journal Applied Spectroscopy. A patent application has been filed. We developed a new method for determining muscle oxygen saturation (SmO2) and another new method for calculating muscle PO2 from SmO2 and the hemoglobin dissociation curve. A paper has been submitted and a patent application filed. We independently validated the PO2 measurement by showing that it was linearly related to stroke volume and inversely related to total peripheral resistance in an LBNP study simulating hemorrhagic shock. A paper on these results has been submitted. We independently validated the pH measurement by showing that anaerobic threshold calculated from hydrogen ion concentration (derived from pH) was highly correlated with the lactate threshold during a VO2max test. This paper is currently being written. 2) Leg System. A system for use on the leg during cycling was completed and delivered to NASA-JSC Exercise Physiology Lab. The leg system with a 30 mm sensor was tested on 10 subjects doing an VO2max cycling test and 2 subjects in a walk/run treadmill test. The 30 mm sensor could be used because these fit subjects had thin layers of fat over their thigh muscle. A special sensor holder was designed to hold the sensor against the skin and completely eliminated motion artifacts from spectra during cycling, walking and running. 3) Trauma Application. We completed enough subjects to demonstrate that in patients with severe sepsis SmO2 was highly correlated with blood lactate. This very important finding provides a noninvasive method for assessing severity of illness and adequacy of resuscitation. SmO2 is an indicator of microvascular perfusion and low SmO2 indicates very poor blood flow, inadequate oxygen delivery, and consequently production of lactate. We determined on a preliminary basis that SmO2 of 40% corresponds to lactate of 4 mmol/l and suggest that a treatment goal might be to resuscitate to

	assure that SmO2 is significantly greater than 40%. One of our prototype systems has been supplied to a collaborator at Beth Israel Deaconess Hospital in Boston who will collect additional data for this study data.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Abstracts for Journals and Proceedings	Soller BR, Soyemi OO, Landry MR, Yang Y, Walz JM, Scott P, Heard SO, Mullen M. "Noninvasive muscle oxygen saturation is correlated with lactate, not ScvO2 in septic patients." Society of Critical Care Medicine, Orlando, FL, December, 2006. Crit Care Med. 2006;34:A54. , Dec-2006
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Articles in Peer-reviewed Journals	Soller BR, Hagan RD, Shear M, Walz JM, Landry M, Anunciacion D, Orquiola A, Heard SO. "Comparison of intramuscular and venous blood pH, PCO(2) and PO(2) during rhythmic handgrip exercise." Physiol Meas. 2007 Jun;28(6):639-49. PMID: 17664618 , Jun-2007
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Awards	Soller, BR, Soyemi, OO, Landry, MR, Yang, Y, Walz, JM, Scott, P, Heard, SO, Mullen, M. "Research Citation Finalist, Society of Critical Care Medicine, Feb 2007." Feb-2007
Patents	Application 60/809,238. Patent, 03/2007. Mar-2007 Soyemi OO, Soller BR. "Method for Calculating Tissue Oxygen Saturation from Optical Reflectance Spectra."
Patents	Patent no. 60/674,379. Patent Pending, April 2006. Apr-2006 Yang, Y, Soller, B R, Soyemi, O, Shear, M. "Systems and Methods for Correcting Optical Reflectance Measurements."